सहायक प्राध्यापक, शासकीय औषधनिर्माण महाविद्यालय,
महाराष्ट्र अभियांत्रिकी शिक्षक सेवा, गट-अ
उच्च व तंत्रशिक्षण विभाग
Assistant Professor in Government Pharmacy Colleges,
Maharashtra Engineering Teachers' Services, Gr.-A
Higher and Technical Education Department

परीक्षेचे टप्पे - लेखी परीक्षा - २०० गुण, **मुलाखत** - ५० गुण.

-: परीक्षा योजना :-

विषय व संकेतांक	प्रश्नसंख्या	गुण	कालावधी	दर्जा	माध्यम	प्रश्नपत्रिकेचे स्वरुप
विहित अभ्यासक्रमावर आधारीत (११०८)	१००	२००	एक तास	पदवी	इंग्रजी	वस्तुनिष्ठ बहुपर्यायी

अ) नकारात्मक गुणदान -

१) प्रत्येक चुकीच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/ कमी करण्यात येतील.
२) एखाद्या प्रश्नाची एकापेक्षा अधिक उत्तरे दिली असल्यास अथवा ज्या उमेदवाराने उत्तरपत्रिकेत पूर्ण वर्तुळ चिन्हांकित केले नसेल अशा प्रश्नाचे उत्तर चुकीचे समजण्यात येऊन त्या प्रश्नाच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/कमी करण्यात येतील.
३) वरीलप्रमाणे कार्यपध्दतीचा अवलंब करताना एकूण अंतिम गुणांची बेरीज अपूर्णांकात आली तरीही ती अपूर्णांकातच राहील व पुढील कार्यवाही त्याच्या आधारे करण्यात येईल.
४) एखाद्या प्रश्नाचे उत्तर अनुत्तरित असेल तर, अशा प्रकरणी नकारात्मक गुणांची पध्दत लागू असणार नाही.

ब) अंतिम गुणवत्ता यादी ही वस्तुनिष्ठ परीक्षेतील व मुलाखतीतील एकत्रित गुणांवर आधारीत राहील.

-: अभ्यासक्रम :-

खालील घटक व उपघटकांचा समावेश असेल.

Sr. No.	Topics and sub-topics
1)	**PHARMACEUTICS**
	Historical background and development of profession of pharmacy, Pharmacopoeias, IP, BP, USP and Extra Pharmacopoeia. Prescription, Posology, Pharmaceutical calculations, **Dosage forms-** Powders, Liquid dosage forms, Monophasic liquids, Biphasic liquids, aerosols, Suppositories, Semisolid dosage forms, solid dosage forms- tablet, capsule, pellets. Sterile dosage forms-parenteral, ophthalmic Cosmetics Pharmaceutical incompatibilities, Solubility of drugs, States of Matter and properties of matter, Physicochemical properties of drug molecules, Surface and interfacial phenomenon, Complexation and protein binding, pH, buffers and Isotonic solutions, Sorensen's pH scale, pH determination (electrometric and calorimetric), applications of buffers, buffer equation, buffer capacity, buffers in pharmaceutical and biological systems, buffered isotonic solutions. **Classification of dispersed systems and their general characteristics**- Colloidal, coarse. Rheology, Deformation of solids, Micromeretics, Drug stability, Preformulation Studies for all types of formulations. Packaging Materials Science. Biopharmaceutics and Pharmakokinetics- Bioavailability and Bioequivalence. Pilot plant scale up techniques. **Novel drug delivery systems**-Like Controlled, Polymers, Microencapsulation, nano, Mucosal, Implantable, Transdermal, Gastroretentive, Nasopulmonary, Ocular, Intrauterine.

2)	**PHARMACEUTICAL CHEMISTRY**
	Inorganic- General methods of preparation & assay, Acids and Bases, Major extra and intracellular electrolytes, Dental products, **Gastrointestinal agents**-Acidifiers; Antacid; Cathartics; Emetics, Antimicrobials, Expectorants, Haematinics, Astringents, Radiopharmaceuticals. **Organic**-Classification, nomenclature and isomerism. General methods of preparation and reactions of organic compounds like Alkanes, Alkenes and Conjugated dienes, Alkylhalides, Alcohols, Carbonyl compounds, Carboxylic acids, Aliphatic amines, Benzene and its derivatives, Phenols, Aromatic Amines, Aromatic Acids, Fatty acids & their reactions, Polynuclear hydrocarbons, Cyclo alkanes. **Stereo isomerism**- Optical isomerism, Geometrical isomerism. Heterocyclic compounds- Nomenclature and classification Synthesis, reactions and medicinal uses. **Biochemistry**-Bioenergetics, Biomolecules- Carbohydrate, lipid, Amino acid, Nucleic acid their role and metabolism, Biological oxidation, Enzymes **Medicinal Chemistry**- Study of the following classes of drugs, Classification, mechanism of action, uses of drugs mentioned in the course (B Pharm syllabi prescribed by PCI), Structure activity relationship of selective class of drugs and synthesis of drugs acting on ANS, CNS, Antihistaminic agents, GIT, Cancer, endocrine, CVS, local anaesthetics, chemotherapetutic agents. Drug design. Combinatorial Chemistry.
3)	**PHARMACEUTICAL ANALYSIS and QUALITY ASSURANCE**
	Pharmaceutical analysis- Different techniques of analysis, Primary and secondary standards. Errors,Sources of errors, types of errors, methods of minimizing errors,accuracy, precision and significant figures Pharmacopoeia, Sources of impurities in medicinal agents, limit tests. **Instrumental Methods of analysis - Theory, Instrumentations and Applications of-**UV Visible spectroscopy, Fluorimetry, IR spectroscopy, Flame Photometry, Atomic absorption, Conductometry, Potentiometry, Polarography. **Chromatography-** Adsorption and partition column chromatography, Thin layer chromatography, Paper chromatography, Electrophoresis, Gas chromatography, High performance liquid chromatography (HPLC), Ion exchange chromatography, Gel chromatography, Affinity chromatography. **Quality Assurance and Quality Management concepts-** Definition, elements, philosophies**,** Quality Management (TQM), Quality management systems. ICH Guidelines, QbD, NABL accreditation, Equipments and raw materials, Personnel. Good Laboratory Practices. Document maintenance in pharmaceutical industry, Calibration and Validation, Warehousing, Technology development and transfer. Regulatory affairs, Regulatory requirements for drug approval, Indian Regulatory Requirements.
4)	**PHARMACOLOGY**
	Human Anatomy and Physiology- Definition and scope of anatomy and physiology, levels of structural organization and body systems, basic life processes, homeostasis, basic anatomical terminology.Cellular and Tissue level of organization **Anatomy and Physiology of following systems-**Integumentary, Skeletal, Osseus and joints, hematological (blood), Lymphatic, cardiovascular, Digestive, Respiratory, Urinary, Endocrine, Reproductive, Nervous **(**Organization of nervous system, neuron, neuroglia, classification and properties of nerve fibre, electrophysiology, action potential, nerve impulse, receptors, synapse, neurotransmitters)- CNS, PNS (including ANS). Special Senses. **Genetics-** Chromosomes, genes and DNA, protein synthesis, genetic pattern of inheritance **Pharmacology-**A. **General Pharmacology- i.** Definition, historical landmarks and scope of pharmacology, nature and source of drugs and routes of drug administration, **ii.** Pharmacokinetics, Pharmacodynamics, factors modifying drug action, Bioassays, Chronopharmacology, Principles of toxicology. B. **Pharmacology** (the term Pharmacology used here refers to the classification, pharmacokinetics, pharmacodynamics-mechanism of action, adverse effects, contraindications, Therapeutic uses, dose and route of administration along with interactions) **of drugs acting on following systems / diseases / disorders-** Peripheral nervous system, Central nervous system, Cardio Vascular System, Urinary system, Autocoids and related drugs, Endocrine system, Respiratory

	system, Gastrointestinal Tract, **Chemotherapy**- (Against microbes, protozoa, metazoan, fungi, viruses, cancer).
5)	**PHARMACY PRACTICE**
	Hospital Pharmacy- Hospital, Hospital pharmacy and their organization, Hospital formulary Drug distribution system in a hospital, Education and training program in the hospital. Adverse drug reaction, Therapeutic drug monitoring Community Pharmacy, Community pharmacy management, Medication adherence, Patient medication history interview Patient counseling. Pharmacy and therapeutic committee. Drug information services. Prescribed medication order and communication skills. Clinical Pharmacy. Interpretation of Clinical Laboratory Tests. Sociology and health, Hygiene and health, Preventive medicine. -National health programs (objectives, functioning and outcome), Community services in rural, urban and school health, **PATHOPHYSIOLOGY (Etiology, symptoms, diagnostic tests) of following diseases/ disorders-** Basic principles of Cell injury, Adaptation, inflammation and repair. **Cardiovascular System-** Hypertension, congestive heart failure, ischemic heart disease (angina,myocardial infarction, atherosclerosis and arteriosclerosis), **Respiratory system-**Asthma, Chronic obstructive airways diseases. **Renal system**- Acute and chronic renal failure, **Haematological Diseases-** Iron deficiency, megaloblasticanemia (Vit B12 and folic acid), sickle cell anemia, thalasemia, hereditary acquired anemia. **Endocrine system**,Diabetes, thyroid diseases, disorders of sex hormones, **Nervous system**,Epilepsy, Parkinson's disease, stroke.**Psychiatric disorders-**depression, schizophrenia and Alzheimer's disease.**Gastrointestinal system**,Peptic Ulcer, Inflammatory bowel diseases, jaundice, hepatitis (A,B,C,D,E,F), Alcoholic liver disease. **Disease of bones and joints-**Rheumatoid arthritis, osteoporosis and gout. **Cancer**- classification, etiology and pathogenesis of cancer, **Infectious diseases-**Meningitis, Typhoid, Leprosy, Tuberculosis Urinary tract infections. Sexually transmitted diseases, AIDS, Syphilis, Gonorrhea Essential drugs concept, National List of Essential Medicines (NLEM)
6)	**PHARMACOGNOSY**
	Introduction, Definition, history, scope and development, Sources of Drugs , Organized drugs, unorganized drugs. Classification of drugs, Quality control of Drugs of Natural Origin, Conservation of medicinal plants, -Plant tissue culture, Edible vaccines. Pharmacognosy in various systems of medicine. Introduction to secondary metabolites. **Study of biological source, chemical nature and uses of drugs of natural origin containing following constituents-**(General introduction, detailed study with respect to chemistry, sources, preparation, evaluation, preservation, storage, therapeutic uses and commercial utility as Pharmaceutical Aids and/or Medicines)- **-Plant Products, Fibers** - Hallucinogens, Teratogens, Natural allergens, **-Carbohydrates, Proteins and Enzymes –Lipids, Marine Drugs, Alkaloids,** Phenylpropanoids and Flavonoids, Steroids, Cardiac Glycosides & Triterpenoids, Volatile oils, Tannins, Resins, Glycosides, Iridoids, Other terpenoids & Naphthaquinones, Biopesticides/Bioinsecticides, Neutraceuticals, Herbal Cosmetics, Herbal excipients, Herbal formulations. **Indian Systems of Medicine.Herbal-Drug and Herb-Food Interactions,** **Evaluation of Drugs** WHO & ICH guidelines for the assessment of herbal drugs Stability testing of herbal drugs.Patenting and Regulatory requirements of natural products. Definition of the terms- Farmers right, Breeder's right, Bioprospecting and Biopiracy, Patenting aspects of Traditional Knowledge and Natural Products. **Regulatory Issues** - Regulations in India (ASU DTAB, ASU DCC), Regulation of manufacture of ASU drugs - Schedule Z of Drugs & Cosmetics Act for ASU drugs
7)	**ALLIED SUBJECTS**
	COMPUTER APPLICATIONS **Concept of Information Systems and Software , I**nformation gathering, requirement and feasibility analysis, data flow diagrams, process specifications, input/output design, process life cycle, planning and managing the project. Introduction to databases, MYSQL, MS ACCESS, Pharmacy Drug database

	Application of computers in Pharmacy –Drug information storage and retrieval, Pharmacokinetics, Mathematical model in Drug design, Hospital and Clinical Pharmacy, Electronic Prescribing and discharge (EP) systems, barcode medicine identification and automated dispensing of drugs, mobile technology and adherencemonitoring. Diagnostic System, Lab-diagnostic System, Patient Monitoring System,Pharma Information System Bioinformatics, Introduction, Objective of Bioinformatics, Bioinformatics Databases, Concept of Bioinformatics, Impact of Bioinformatics in Vaccine Discovery Computers as data analysis in Preclinical development, Chromatographic data analysis (CDS), Laboratory Information management System (LIMS) and Text Information Management System (TIMS) **ENVIRONMENTAL SCIENCES-** Renewable and non-renewable resources, Natural resources and associated problems, -Ecosystems Environmental Pollution, Air pollution; Water pollution; Soil pollution **MICROBIOLOGY-** Introduction- Study of ultra-structure and morphological classification of bacteria, Fungi and Virus, nutritional requirements, raw materials used for culture media and physical parameters for growth, growth curve, isolation and preservation methods for pure cultures, cultivation of anaerobes, quantitative measurement of bacterial growth (and viable count). Identification of bacteria using staining techniques, methods of sterilization, Evaluation of the efficiency of sterilization method, Sterility indicators, Sterility testing of products, disinfection and disinfectants. Microbiological assay. Growth of animal cells in culture, general procedure for cell culture, Primary, established and transformed cell cultures. Application of cell cultures in pharmaceutical industry and research. **BIOTECHNOLOGY-** a) introduction, Enzyme Biotechnology, Biosensors- Protein Engineering., Use of microbes in industry, Production of Enzymes-genetic engineering.cloning vectors, Recombinant DNA technology. Application of r DNA technology and genetic engineering in the products, Interferon b) Vaccines- hepatitis- B c) Hormones- Insulin. Types of immunity- humoral immunity, cellular immunity, vaccines and other products relative to immunity- their Production, Storage conditions and stability Fermentation methods and general requirements, Blood product collection, Processing and storage of whole volume blood, dried human plasma, plasma substituents- **PHARMACEUTICAL ENGINEERING** Flow of fluids, Size Reduction, Size Separation-Evaporation, Heat Transfer, Distillation, Drying, Mixing, Filtration, Centrifugation, Materials of pharmaceutical plant construction, Corrosion and its prevention.
8)	**BIOSTATISTICS, RESEARCH METHODOLOGY AND PHARMACEUTICAL JURISPRUDENCE**
	BIOSTATISTICS- Introduction,, Measures of central tendency, Measures of dispersion, Correlation, Regression, Probability, Parametric test, - Non Parametric tests, Regression modeling, Sample size determination and Power of a study,Statistical Analysis Using Excel, SPSS, MINITAB, **Graphs** **RESEARCH METHODOLOGY- Introduction** Need for research, Need for design of Experiments, Experiential Design Technique, plagiarism, Designing the methodology**,** Protocol, Cohorts studies, Observational studies, Experimental studies, Designing clinical trial,. Report writing and presentation of data, **Design and Analysis of experiment-** Factorial Design, DESIGN OF EXPERIMENTS Response Surface Methodology, Central composite design, Historical design, Optimization Techniques Research grants and funding agencies like- CSIR, DBT, DST, UGC, ICMR and all such national international Research funding agencies **JURISPRUDENCE-** Drugs and Cosmetics Act, 1940 and its rules 1945, Pharmacy Act –1948, Medicinal and Toilet Preparation Act –1955, Narcotic Drugs and Psychotropic Substances Act-1985 and Rules, Study of Salient Features of Drugs and magic remedies Act and its rules, Prevention of Cruelty to animals Act-

	1960, National Pharmaceutical Pricing Authority **Code of Pharmaceutical ethics -** Definition, Pharmacist in relation to his job, trade, medical profession and his profession, Pharmacist's oath **Right to information Act 2005,** Maharashtra Right to Public Services Act 2015 **Introduction to Intellectual Property Rights (IPR)-**Patent, trademark, copyright.
9)	**IMPORTANT SERVICE RULES AND ADMINISTRATIVE PROCESSES**
	Rules and orders applicable in relation to administrative work , Maharashtra Civil Services (General Conditions of Service) Rules 1981 Maharashtra Civil Services (Pay) Rules 1981 Maharashtra Civil Services (Provisions during Period of Joining, Foreign Service, Suspension, Dismissal from Service) Rules 1981 Maharashtra Civil Services (Leave) Rules 1981 Maharashtra Civil Services (Pension) Rules 1981 Maharashtra Civil Services (Discipline and Appeal) Rules 1979 Departmental Inquiry Book Maharashtra Civil Services (Conduct) Rules 1979 Purchase procedure- Financial authorities and working of GeM portal **Various scholarships under MAHADBT their rules regulations as per GR** **Admission rules / regulations for Pharmacy admissions** (Excluding Diploma), process of admissions, activities to be done by FC, SC, admitting college as per CET cell notifications. Accreditation processes like- NBA, NAAC, IQAC National Education Policy (NEP). Various desks at DTE and their working
10)	**INTELLIGENCE TEST**
	Questions will be asked to test how far candidates can think quickly and accurately.

Important Note –

1) The basic Pharmacy syllabus (Serial Number 1 to 8) is based on the syllabi prescribed by Pharmacy Council of India for B Pharm course, for any clarifications / additional details- the syllabus prescribed by PCI should be referred to and treated as final. The topics in **bold letters** indicate major area having sub topics.
2) Wherever Acts / Rules are prescribed in above syllabus, latest amendments/versions to be considered.

दिनांक – ११ फेब्रुवारी, २०२५

अवर सचिव
महाराष्ट्र लोकसेवा आयोग